Machine Learning

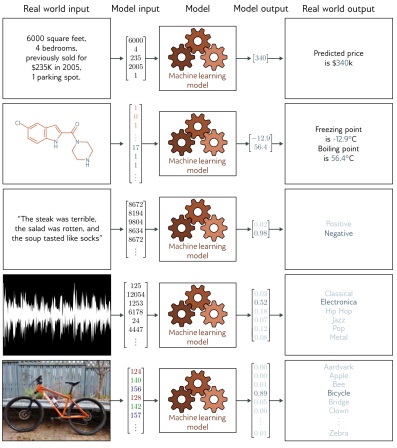
2. Classification, with and beyond the linear model

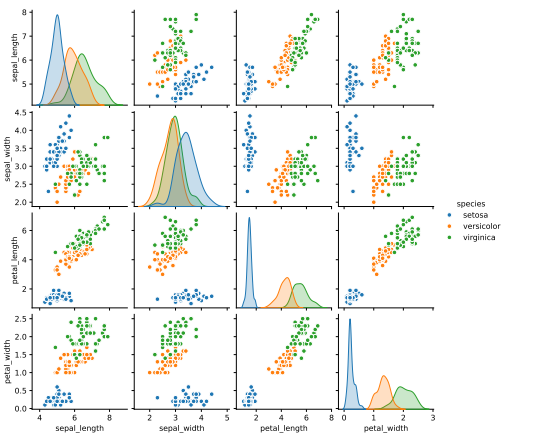
Louis Bagot
Université Claude Bernard Lyon 1

September 28, 2026

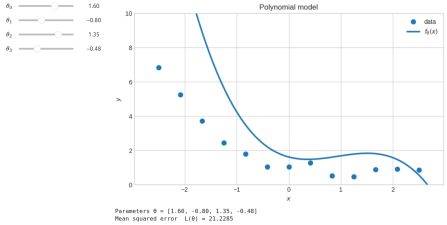
Where were we?

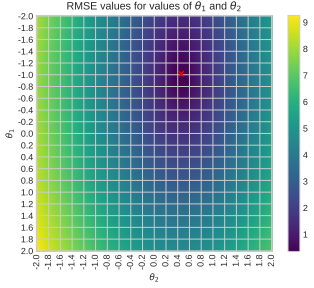
ML: Approximate a function from data examples $\{x_i, y_i\}$

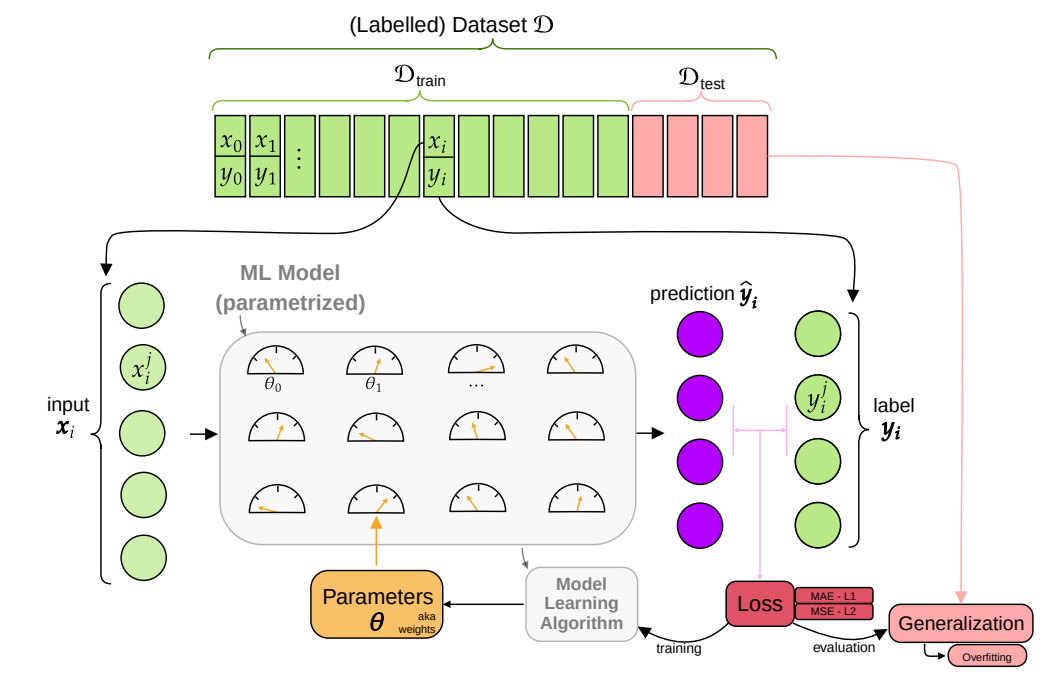




Parametrized models $f_\theta(x_i) \approx y_i$ learn parameters θ to minimize loss: $\arg\min_\theta \mathcal{L}(\mathcal{D}, f_\theta)$






(Labelled) Dataset $\mathcal{D}$
$\mathcal{D}_{\text{train}}$
$\mathcal{D}_{\text{test}}$
x_0 y_0
x_1 y_1
x_i y_i
ML Model
(parametrized)
θ_0
θ_1
...
input
$\boldsymbol{x}_i$
x_i^j
prediction $\hat{\boldsymbol{y}}_i$
y_i^j
label
$\boldsymbol{y}_i$
Parameters
$\boldsymbol{\theta}$
aka
weights
Model
Learning
Algorithm
Loss
MAE - L1
MSE - L2
training
evaluation
Generalization
Overfitting

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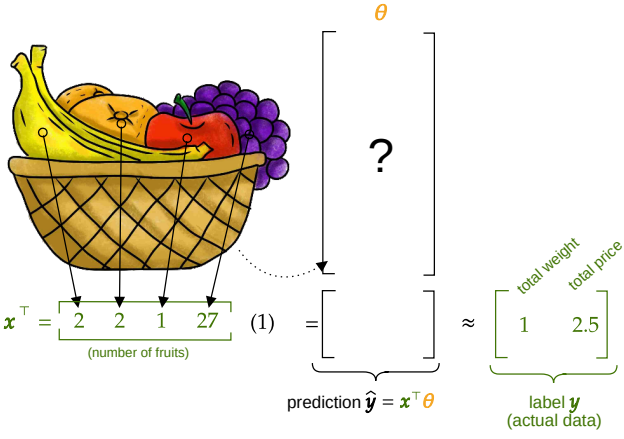
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$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots = \boldsymbol{x}^\top \boldsymbol{\theta}$$

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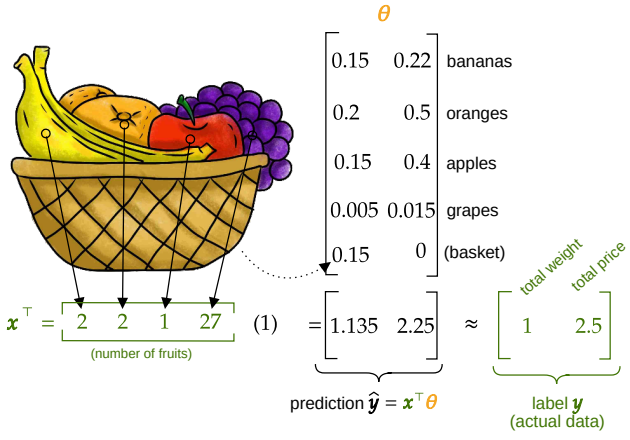
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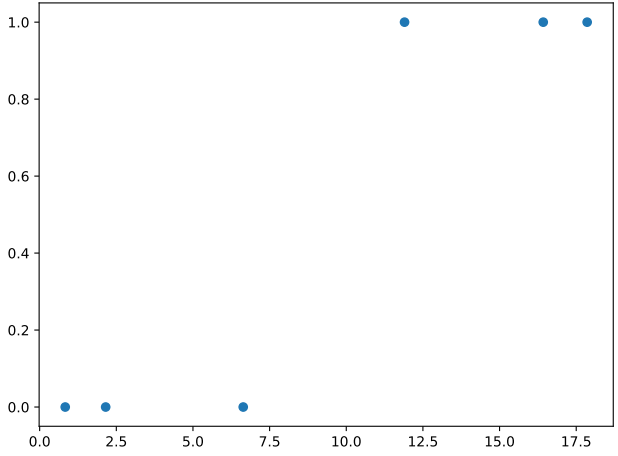
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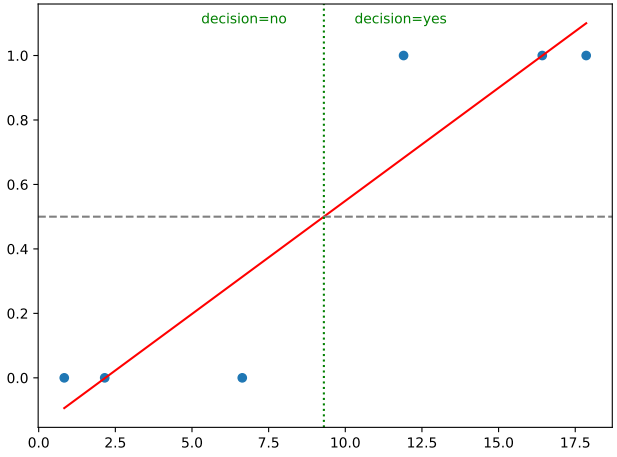
What about classification?

If I need to answer “is this basket more expensive than 10€?” (no:0, yes:1)



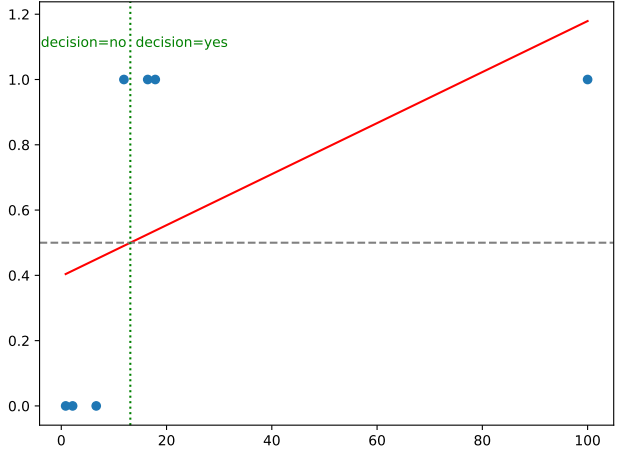
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Naive algo: fit the line and decide “yes” when line > 0.5:



Linear classif.: what if there are outliers?

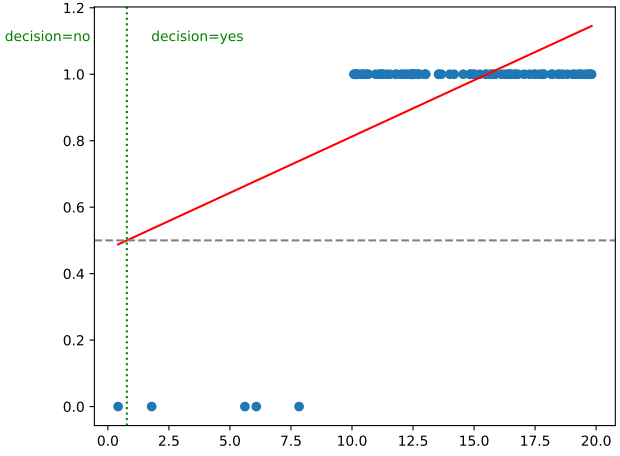
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(Note: MSE & OLS are weak to outliers even outside of classification!)

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- Linear model outputs go way past $[0, 1]$

Outputs in $[0, 1]$: just turn them into probabilities?

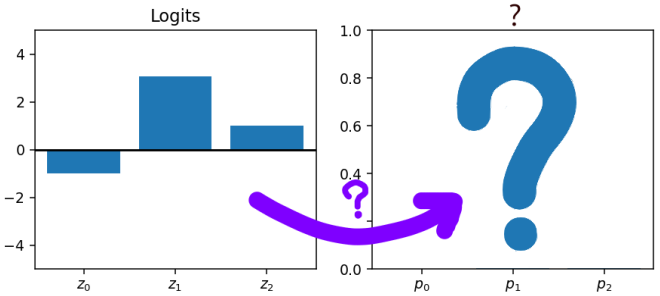
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logit z_c $\xrightarrow{\text{transformation?}}$ $\hat{y}_c = \text{Proba(class c)}$



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First intuition (not great): **linear normalization**.

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- Bad gradient properties: division forces tiny steps

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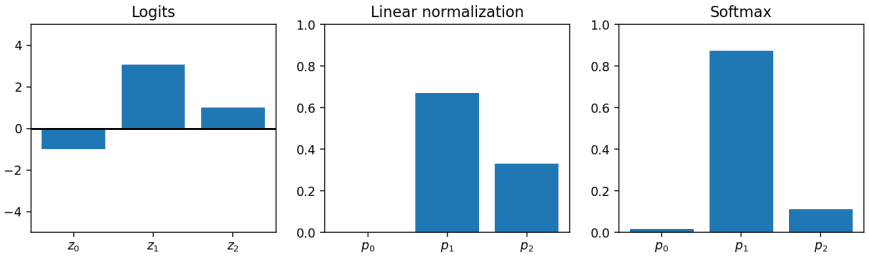
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- Still a division though?

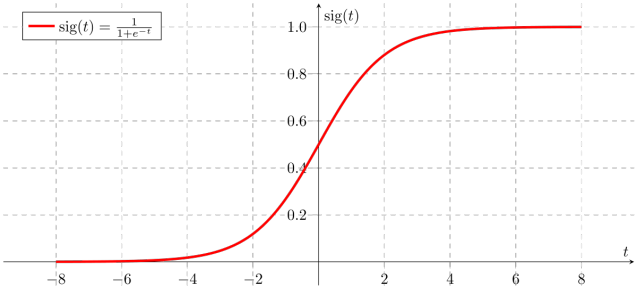
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(you can play with it in the associated interactive notebook)

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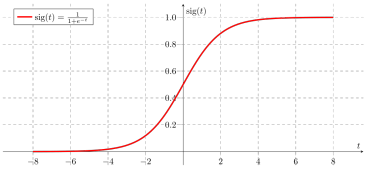


$$\text{sigmoid}(z) \doteq \frac{1}{1+e^{-z}}$$

$$\lim \text{sigmoid}(z) \begin{array}{l} = 0 \text{ as } z \to -\infty \\ = 0.5 \text{ as } z = 0 \\ = 1 \text{ as } z \to +\infty \end{array}$$

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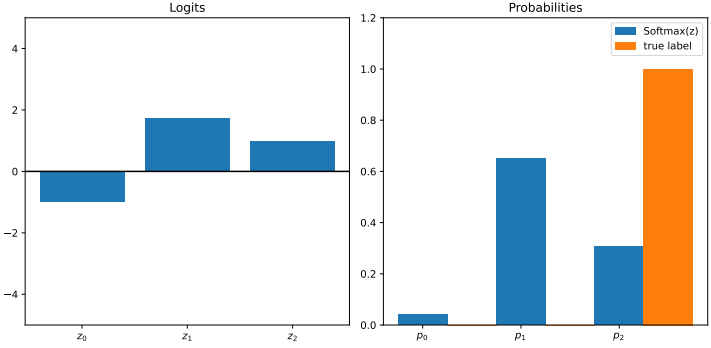


$$\text{Proba(yes)} = \text{sigmoid}(z) \doteq \frac{1}{1+e^{-z}} \iff \log \frac{\text{Proba(yes)}}{\text{Proba(no)}} = \theta_0 + x_1\theta_1 + x_2\theta_2 + \dots$$

$\rightarrow$ an increase of x_i by $+1$ multiplies the **odds ratio** by e^{θ_i}

- multiplications in probas turn into additions in $\log$ probas!

What loss to use with softmax?



We need a good notion of distance between the softmax and the true label.

Distance between distributions?

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Use **logarithms** 😁 through the **Cross-Entropy loss**: for one sample:

$$\mathcal{L}_{\text{CE}}(\hat{\boldsymbol{y}}, \boldsymbol{y}) \doteq -\sum_{\text{class } c=1}^{\text{n_classes}} y_c \log \hat{y}_c$$

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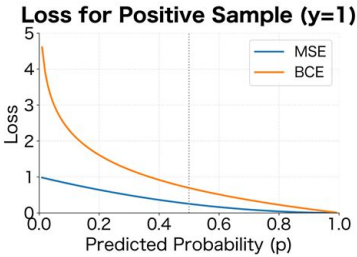
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- Simply aims to **rise the probability of the correct class**.
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- The $\log$ cancels out the softmax $\exp$ and turns the division into a subtraction!

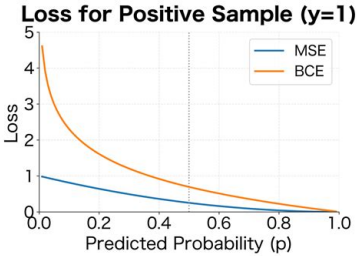
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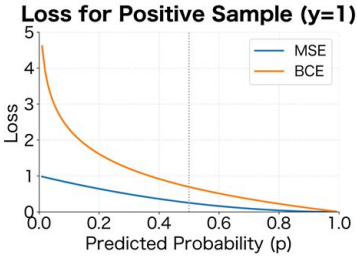
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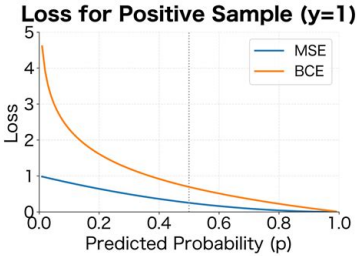
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  - $\to$ this is why the z_i are called $\log$its! $\log$ probabilities!
- As a differentiable function: $\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_c} = \hat{y}_c - y_c$, just like MSE!
  - if prediction was $\hat{y}_c = 1$ for $y_c = 0$, push logit down (descent), otherwise up
  - recovers nice gradients in logit / log probability space.

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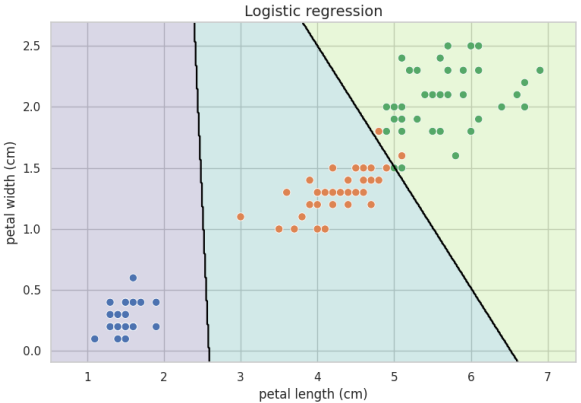
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$\rightarrow$ All this is true **all the way up to Deep Learning!**:

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Classification with Linear models:

$$f_{\theta}(\boldsymbol{x}) = \text{softmax}(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots)$$



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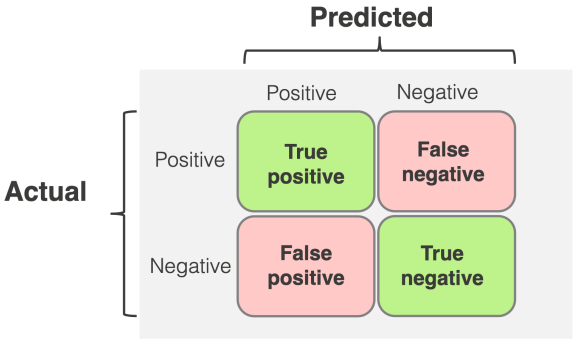
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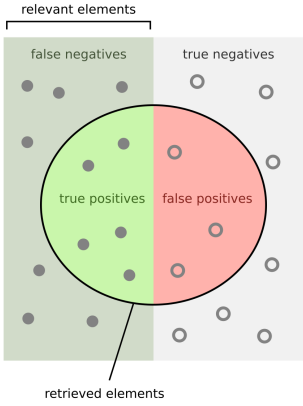
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- Has a huge issue! Can you see it?
  - $\rightarrow$ **Class imbalance**: predict the most frequent class(es), ignore the rest
- (Accuracy is still very useful in many other cases, just be aware!)

Can we improve around the accuracy?

Differentiate between





This is a **confusion matrix**.

Can we improve around the accuracy?

Recall: (probability of detection)

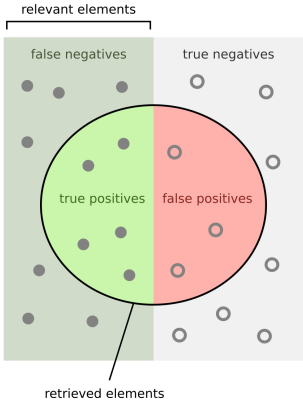
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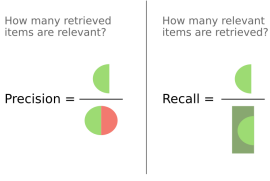
$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision:

Among my model's positives, how much really is?

$$\text{Precision} = \frac{TP}{TP + FP}$$





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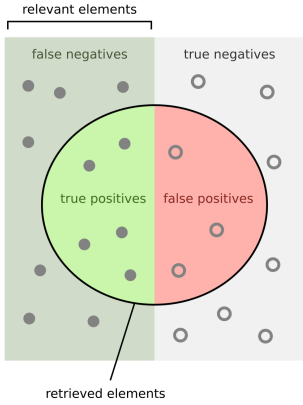
Precision:

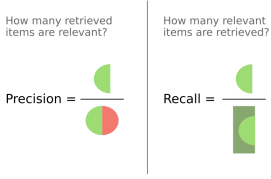
Among my model's positives, how much really is?

$$\text{Precision} = \frac{TP}{TP + FP}$$

F1 Score:

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

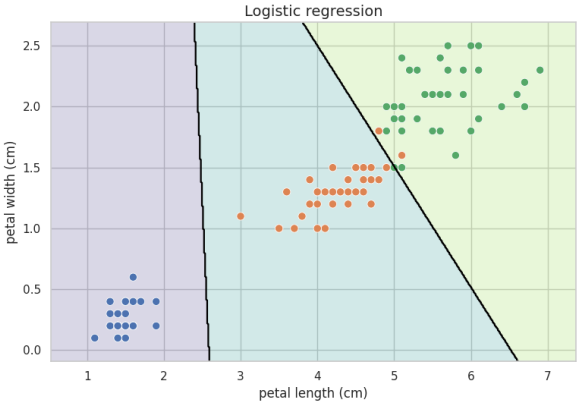




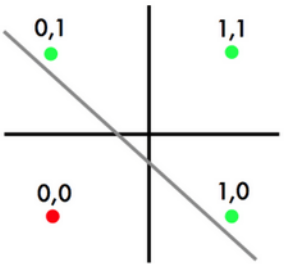
Back to building models!
Can we improve on the linear models?

Linear models:

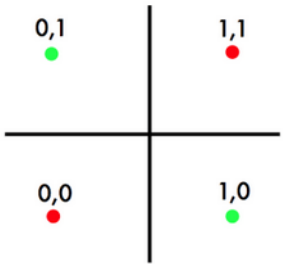
$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$$



What about non-linear relationships?



OR



XOR

What about non-linear relationships?

Many, many families of models (virtually all parametric, $f_{\theta}(\boldsymbol{x})$) with:

- Bunch of parameters θ
- Sequence of computations $f_{\theta}(\boldsymbol{x})$
  - $\rightarrow$ this controls the *shape of the model prediction*
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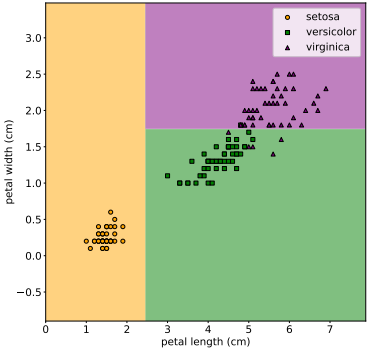
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- Algorithm for learning the parameters (minimizing the loss)

Ideas of non-linear models?

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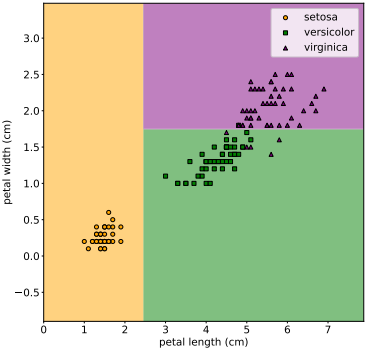
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This is a **Decision Tree**!



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- **Where** do we split?
  - Consider all features d (data dimensions)
  - Possible split between each sample value (mean of both): $N - 1$ tries

  $\rightarrow$ total $d \times (N - 1)$ tests

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Example with classif:
For split class proportions $\boldsymbol{p} = [p_1, \ldots, p_{\text{n_classes}}]$, with $p_i = $ % of class i in split, compute **Gini impurity:**

$$I_G(\boldsymbol{p}) = 1 - \sum_c p_c^2$$

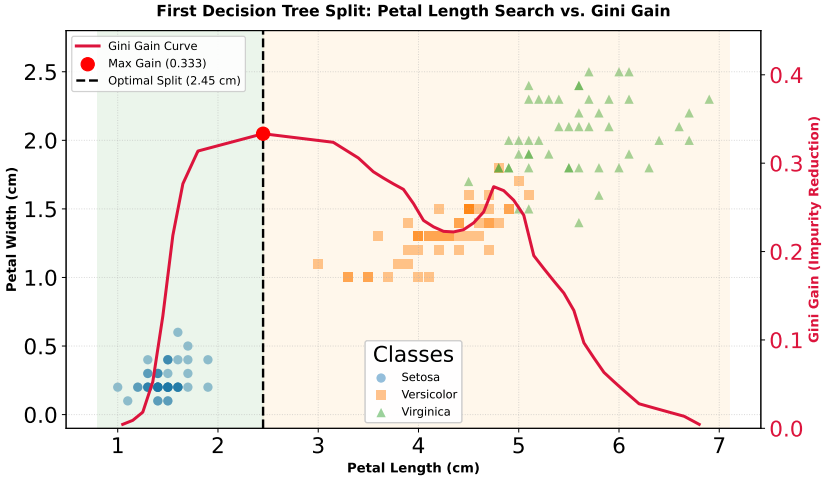
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- Maximal when classes are balanced ("*impure*" split)
- Minimal (=0) at *pure* (single-class) splits.

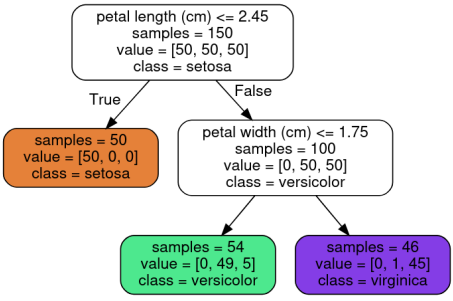
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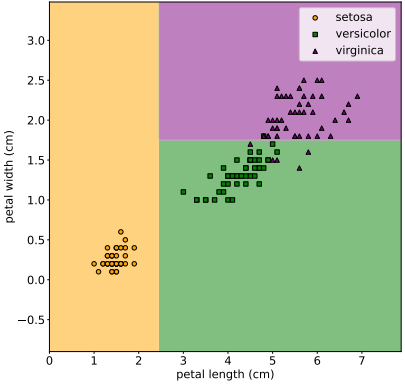


Next split: which part of the space do we split?

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Try both! This is how we get a *tree structure*:





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Training: repeat until target depth / node budget:

- At **each node and feature**, evaluate possible splits
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  - — regression: MSE
  - → weigh each by the number of samples in each side of the split

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Inference: go down the tree for the new input

- classification: most popular class in leaf node
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- Can easily model some non-linear relationships
- Work with little data, and can quickly extract important relationships
- Easily understood and quite explainable: like “expert” systems
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- Limited to splits along the feature directions; often poor performance
- Need to decide a maximum capacity / node or depth count
- Sensitive to little changes in the data (instability)
- Prone to overfitting

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The idea is that **errors cancel out** at scale

- little probability that they all make the same mistakes!

$\longrightarrow$ "wisdom of the crowds"

- This is a first example of **Ensemble Learning!** more info next time :P